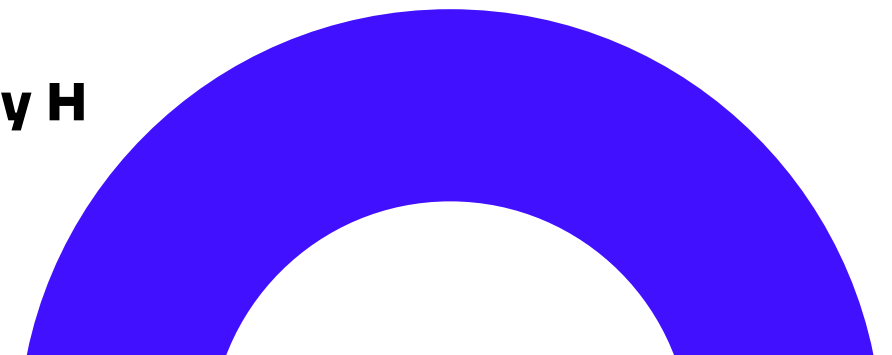


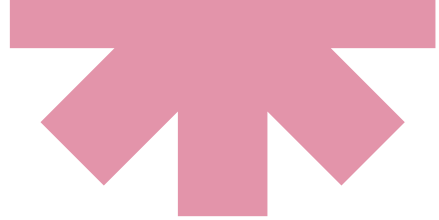
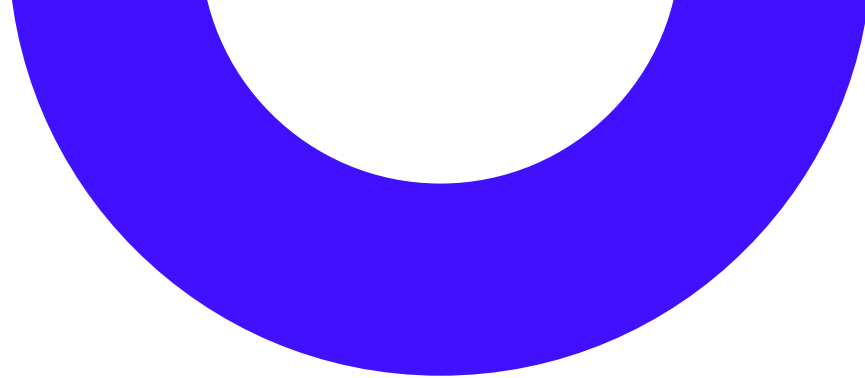
Quantitative Analysis

Execution-Aware Ichimoku + EMA Backtest on Indonesian Commodity Stocks

**Long-only daily trend-following research on PTBA.JK, ANTM.JK,
BRPT.JK, MDKA.JK, and PGAS.JK.**

Presented By: **Yonathan Hary H**

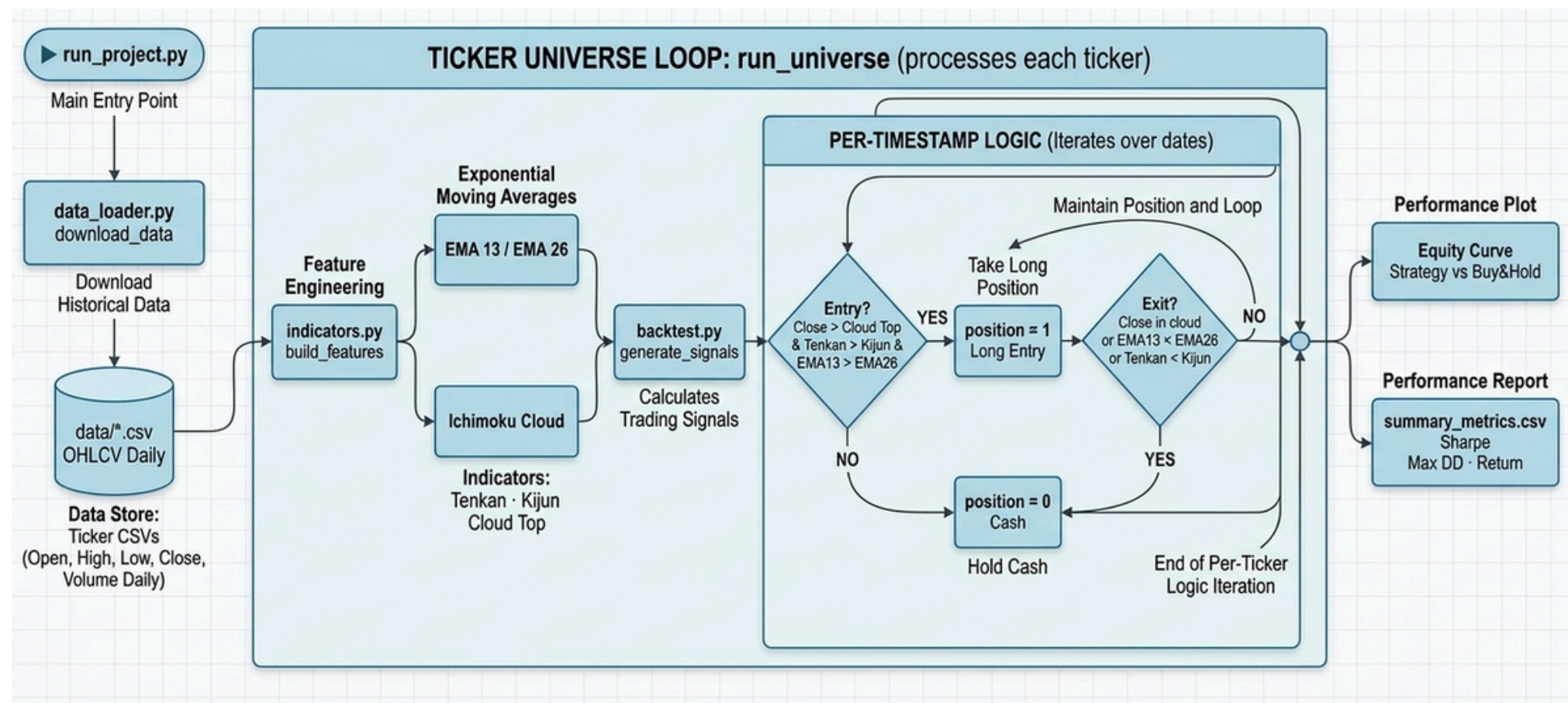




Research Question

**Can a simple trend-following strategy
produce usable risk-adjusted returns after
costs?**

Strategy Rules



Entry

- Close above cloud.
- Tenkan-sen above Kijun-sen.
- EMA 13 above EMA 26.
- Optional volume confirmation.

Exit

- Close falls into or below cloud.
- EMA 13 below EMA 26.
- Tenkan-sen below Kijun-sen.

Data & Universe

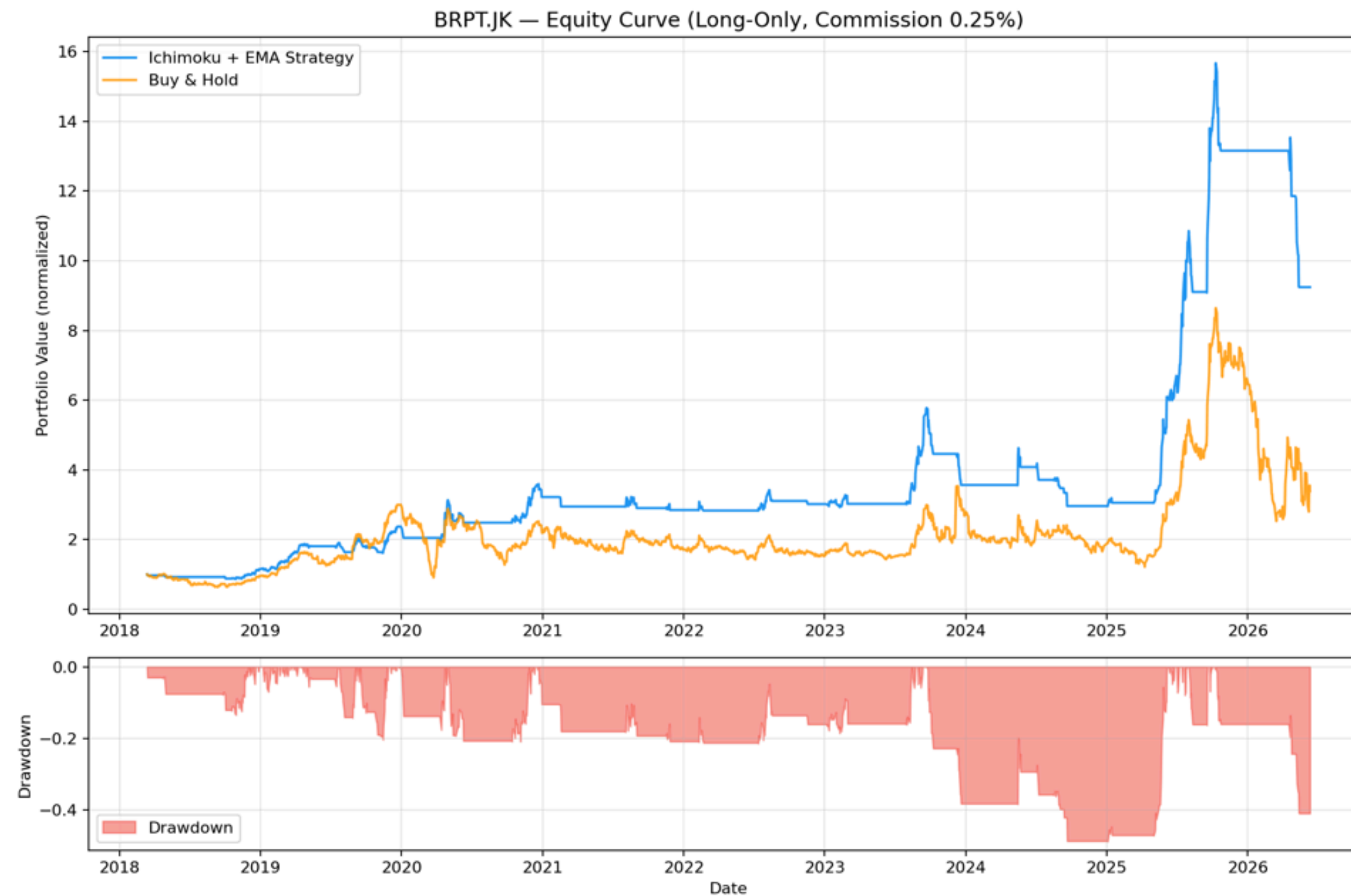
→ Universe



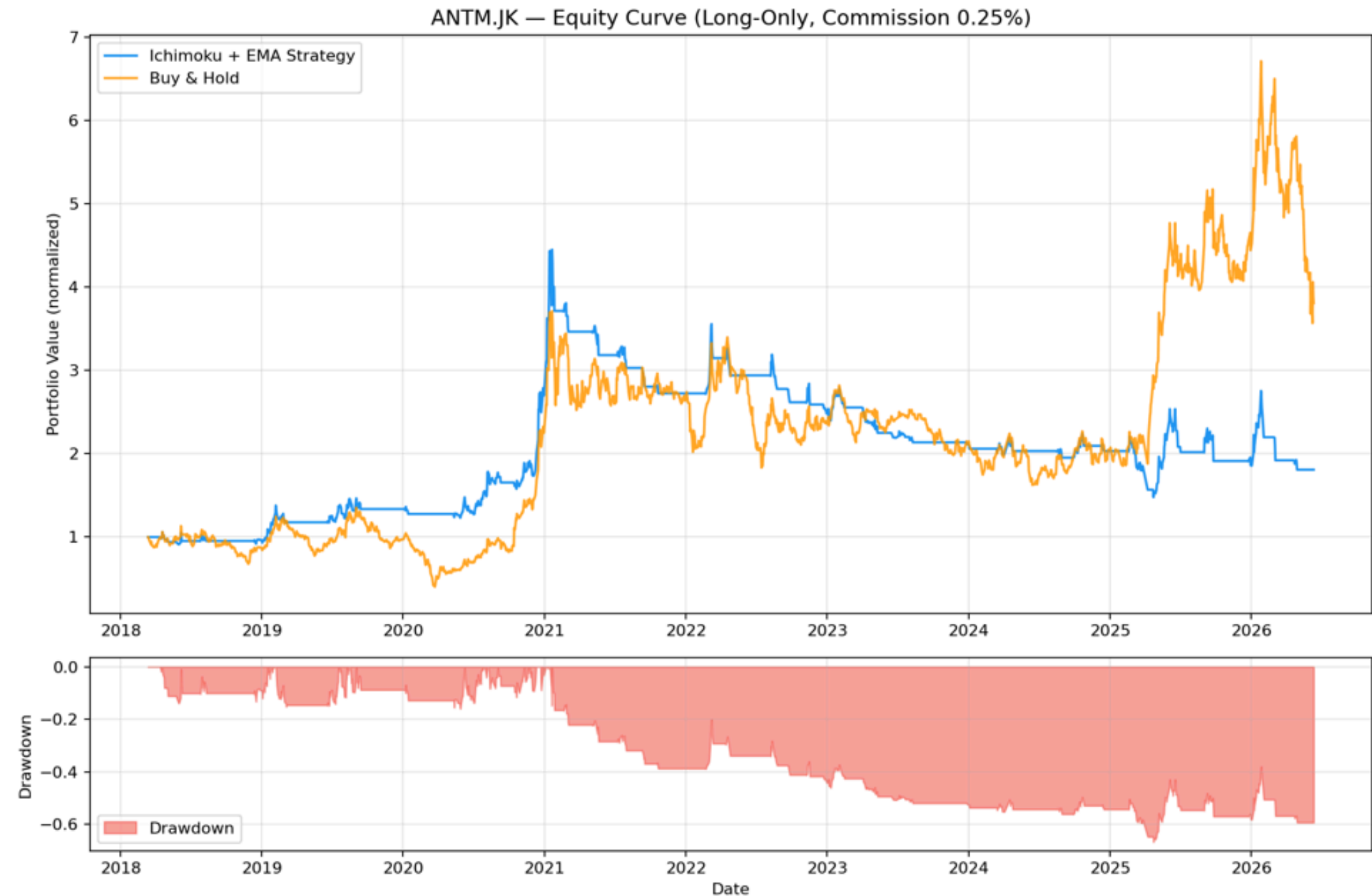
→ Data

- Daily OHLCV Data derived from yfinance from 2018 to 2026
- Long only (behaving like indonesian market)
- Commision: 0.25% (high ceiling for indonesian broker)

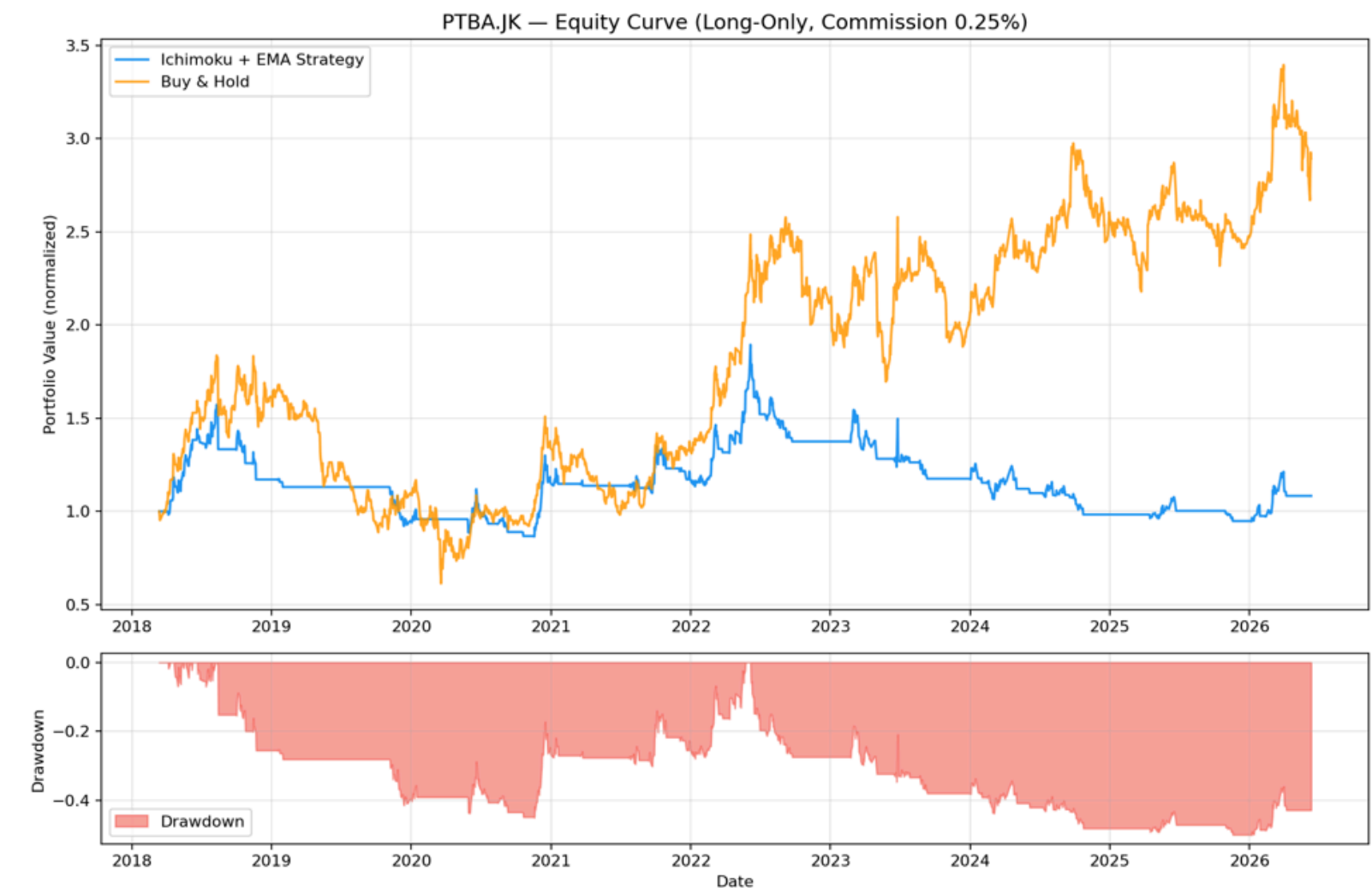
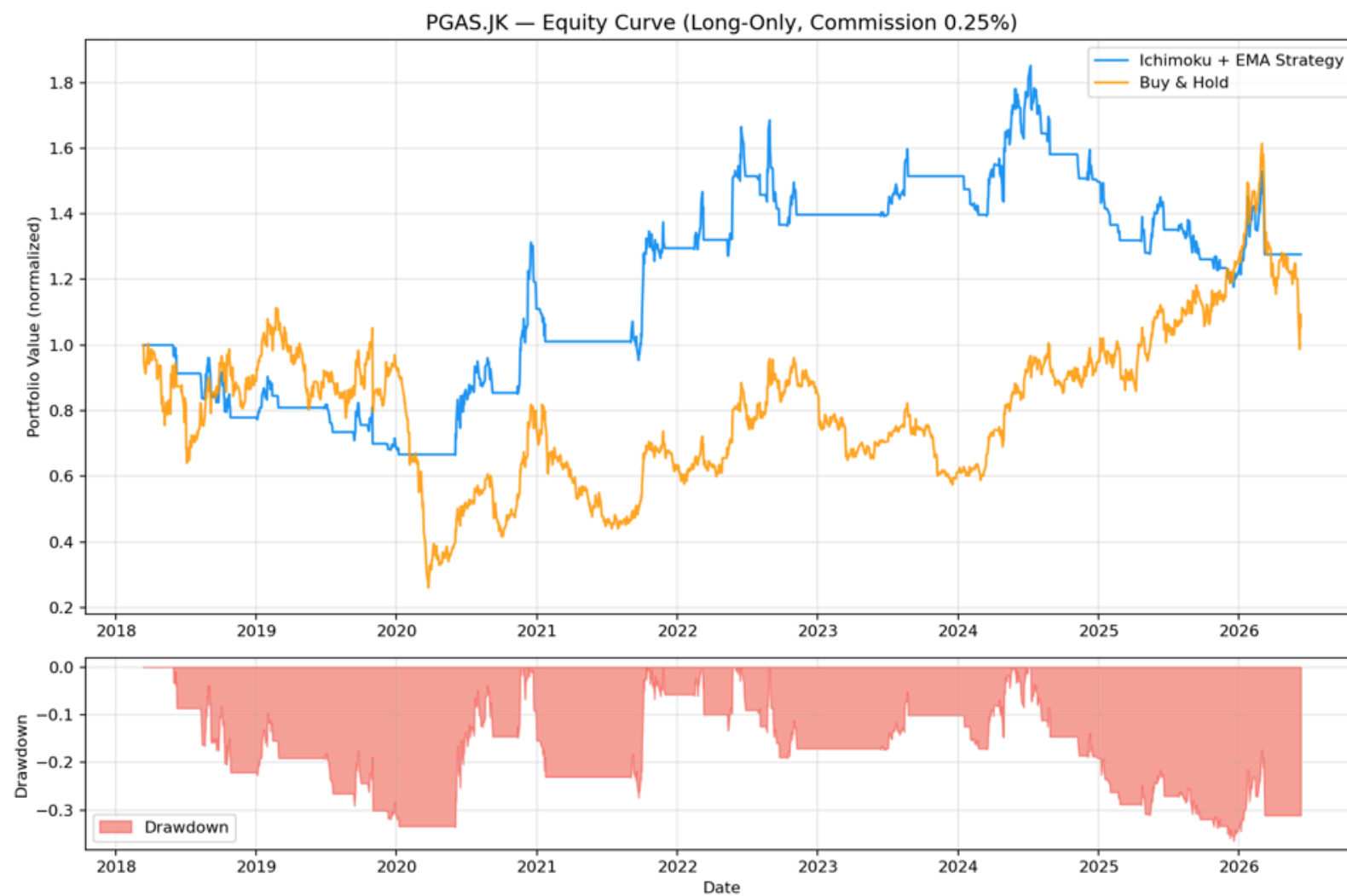
Results – BRPT gave the best return and Sharpe ratio



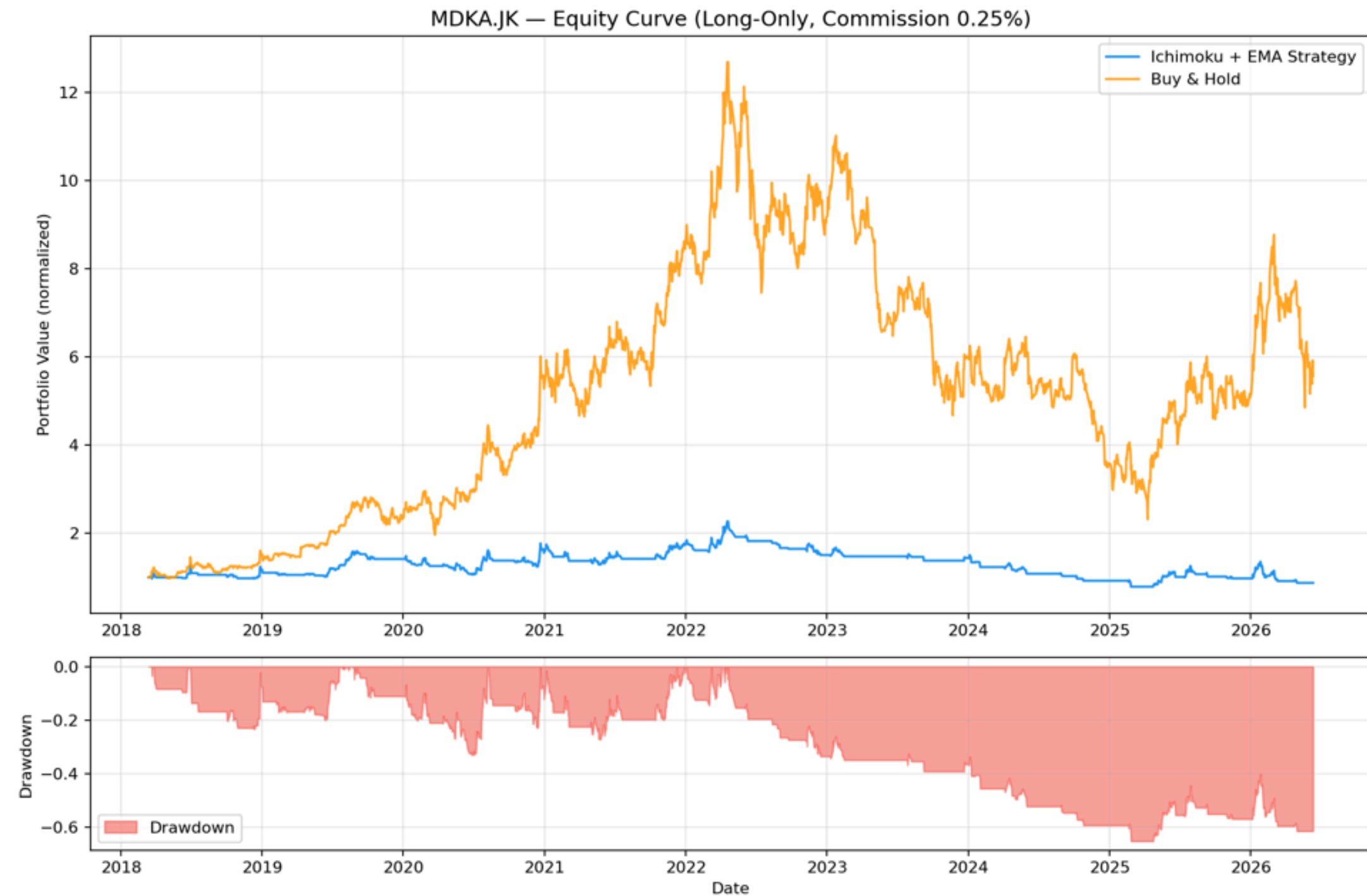
Results – ANTM was positive but weaker than buy-and-hold



Results – PTBA and PGAS were modest



Results – MDKA is underperformed



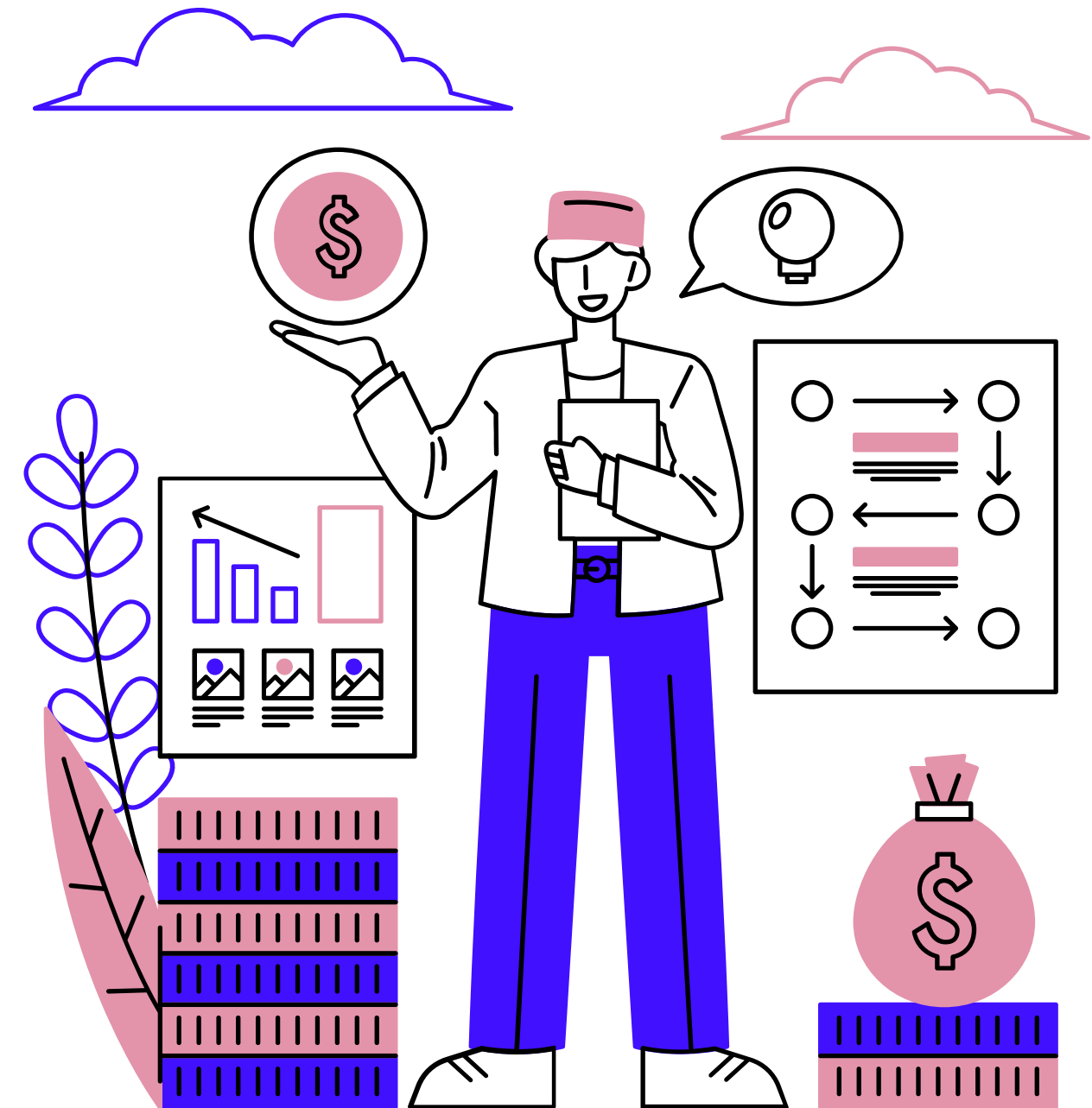


Interpretations

➔ **Understanding what results means**

- The strategy appears to work better in some stocks than others.
- Stronger trend persistence may explain BRPT.JK's better behavior.
- Risk management and exit design matter a lot.

Thank you for listening to my Presentation!



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